SDC HACKATHON 2026 · PHASE 1 PROPOSAL

Timed

AI production planning for small food manufacturers: what to make first, what will run out, and why.

Team TimeOut · Sunway University · Domain 2 · SDG 8

github.com/woopaaaa/Timed---TimeOut

1 · TEAM AND PROJECT OVERVIEW

Who we are

Field	Our entry
Team name	[TimeOut]
Project / solution name	Timed
Selected domain · SDG track	Domain 2 · SDG [8]
Leader · email & phone	Muhammad Husein Ilhamsyah · husein.ilhamsyah@gmail.com
University affiliation(s)	Sunway University
Team members	5
Code repository	github.com/woopaaaa/Timed---TimeOut

2.1 · THE PROBLEM CONTEXT

Small food makers plan by gut feel, and festive rushes punish it

First come, first served

Orders are made in the order they arrive. A big, late-due order can eat the stock that urgent small orders needed.

Stockouts seen too late

Butter, chicken and flour have supplier lead times. By the time a shortage shows, it is too late to reorder.

Festive spikes

Raya, CNY and Deepavali multiply demand for a few weeks. Missed orders and over-production both cost money and food.

Who is affected: owners and staff of small kuih, biscuit and frozen-food makers, and the shops and families who order from them. [Add one cited figure on SME food loss or late deliveries.]

2.2 · THE PROPOSED AI SOLUTION

Four engines, one owner briefing

Demand forecast

Weekly forecast per product with an 80% range; learns festival, Ramadan and school-holiday effects.

Production planner

Earliest-due-date scheduling: each order is on time, late or infeasible, with a reason.

Stockout prediction

The day each material runs out and the last day to reorder, given supplier lead times.

GenAI explanations

An LLM writes the briefing and answers questions in English or Bahasa Melayu.

Workshop skills used: **GenAI APIs** (Groq / Gemini) · **data modelling** (validated inputs, REST API) · **cloud hosting** [platform].

MEASURED ON THE DEMO DATA

Same orders, same stock, more on time

First come, first served

50%

3 of 6 orders on time



Timed plan

67%

4 of 6 orders on time

Butter is short (15 kg for 19.2 kg of cookie orders). Timed makes the two urgent shop orders first, flags the big hamper and raises a butter purchase request for the owner to approve.

FORECAST ACCURACY · 26-WEEK BACKTEST

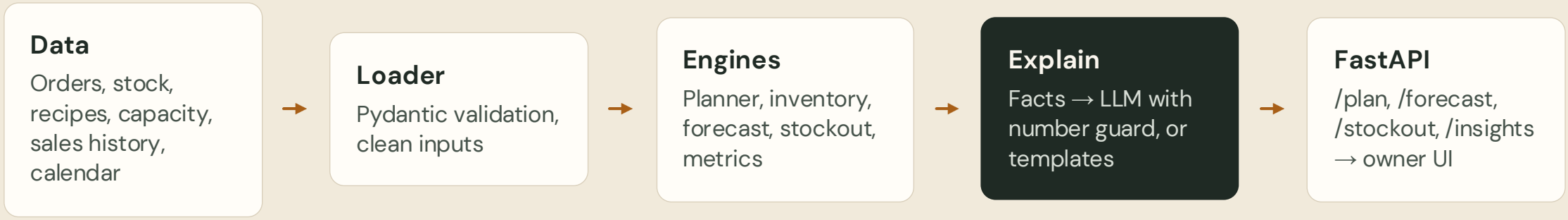
The model beats both simple rules of thumb

Product	Timed model	Last 4 weeks	Same week last year	80% range hit
Butter cookies	6.8%	18.9%	31.2%	77%
Curry puffs	5.3%	13.9%	19.3%	81%
Coconut biscuits	9.2%	20.2%	16.8%	77%

Error is MAPE (lower is better). One log-linear regression per product: trend, price, promo, school holiday, Ramadan and festival-week effects, with outlier removal. It predicts the Deepavali gift-box rush from the calendar.

3.1 · CONCEPTUAL TECH STACK

From order book to owner approval



GenAI API

Groq Llama 3.3 70B, Gemini Flash-Lite or OpenRouter, via OpenAI-compatible HTTP

Cloud hosting

Uvicorn container on [platform / region]; .env secrets never committed

Sponsor resources

[Yes / No: name the SDC sponsor API or cloud credits used]

3.2 · DESIGN THINKING AND DEPLOYMENT

Built around one owner’s Monday morning

Empathise

Owners juggle orders, stock and festive rushes on paper or WhatsApp.

Define

What to make first, what runs out, when to reorder.

Ideate

Calculated numbers, AI only for words; owner keeps the final say.

Prototype

Nine small modules, each with a spec and tests.

Test

113 automated tests pass, no internet or key needed.

Ready for Phase 2

Bilingual (English, Bahasa Melayu) · works offline with template explanations · real data drops in with the same CSV columns · [frontend + hosting plan, user testing with a real SME].

4 · PRELIMINARY SDG IMPACT FRAMEWORK

Our baseline, KPI and Phase 2 target

Metric category	Our response
Target SDG sub-target	[e.g. 12.3 halve food waste and reduce food losses along production chains; or 9.3 small-enterprise access to services]
Current problem baseline	First come, first served delivers 50% of orders on time on the demo data; hand forecasts err 14–31% (MAPE). [Add a cited SME benchmark.]
Core measurable KPI	On-time order rate (%) and forecast error (MAPE %); secondary: kg of ingredients over-produced
Phase 2 impact target	[__]% more orders on time and forecast error under [__]% on a real SME's data

Sources: Timed metrics and 26-week backtest on illustrative data; [SME benchmark citation] · 09

Open-source stack we did not write

Name & version	Purpose in project	License
Python 3	Language for the whole engine	PSF
FastAPI ≥ 0.110	REST API and /docs page	MIT
Uvicorn ≥ 0.29	ASGI web server	BSD-3
Pydantic ≥ 2.6	Input validation and data models	MIT
pandas ≥ 2.1	Loading CSV data	BSD-3
NumPy ≥ 1.26	Forecast regression maths	BSD-3
HTTPX ≥ 0.27	Calling the LLM APIs; API tests	BSD-3
pytest ≥ 8	Automated tests	MIT

From requirements.txt · our own code: Apache 2.0 · 10

TOOLS DECLARATION · 3 APIS · 4 MODELS, DATASETS, AI TOOLS

Every model and dataset we touch

Resource	Type · tier	Used for	License / terms
Groq · Llama 3.3 70B	API, pre-trained · free	Owner briefing, Q&A (EN/BM)	Llama 3.3 licence
Gemini Flash-Lite or OpenRouter	API, pre-trained · free	Backup LLM providers	Provider terms
Timed demand model	Self-trained, NumPy	Weekly forecast, 80% range	Ours, Apache 2.0
Kuih Mesra sample data	Synthetic · no personal data	Orders, stock, sales, events	Ours, Apache 2.0
[Hosting service]	Cloud · [plan / region]	Run the API	[terms]
Claude Code (Anthropic)	AI dev tool · [extent]	Code, specs, slides	Anthropic terms

Review of AI output: every module has a written spec and tests; we read each change and ran pytest before merging. [Add your own words, max 100.]

TOOLS DECLARATION · 6 SECURITY, PRIVACY AND RESPONSIBLE USE

The AI explains; it never decides

Data handling

No personal data in the prototype. Only computed business facts (no customer contacts) go to the LLM. Keys live in a git-ignored .env file. [Retention policy.]

Number guard

Every number in the LLM's text must appear in the calculated facts. If not, the text is discarded and a warning says why.

Human approval

Purchase requests start as pending; only the Owner role approves. The engine never orders anything.

Known limits: free-tier rate limits and single-provider outages fall back to templates with the same numbers; the forecast is only as good as two years of sales history.

5 · 6 · 7 READINESS, IP AND DECLARATION

Original, declared and ready to judge

IP and originality

Question	Answer
Pre-existing work	[Yes / No]
Third-party code, data, models	Yes, all declared
Employment IP conflicts	[Yes / No]
Our licence	Apache 2.0

Checklist

- Mapped to a UN SDG sub-target
- A working application, not a case study
- No credentials in repo or documents
- Generative-AI use disclosed

[Confirm: all members finished the GenAI, Database and Cloud workshops; prototype URL shared.]

THANK YOU

Make the right batch first.

Timed tells a small food maker what to make, what will run out, and why, in their own language.

Code github.com/woopaaaa/Timed---TimeOut